NATIONAL UNIVERSITY OF SINGAPORE

MA1522 – LINEAR ALGEBRA FOR COMPUTING

(Semester 2 : AY2025/26)

SECTION A

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1. Section A contains **FIFTEEN (15)** multiple-choice questions, and each question carries **TWO (2)** marks.
2. This is an open-book examination, and students may use **offline** calculators.
3. This section will be imported into Examplify.

Question 1 [2 marks]

Which of the following matrices are elementary matrices?

(I) $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ (II) $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ (III) $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ (IV) $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$

(A) Only (I) and (II). (B) Only (I) and (III).
(C) Only (I), (II) and (III). (D) None of the above. (D)

Question 2 [2 marks]

Consider a linear system with m equations and n variables whose coefficient matrix (or augmented matrix) has rank m. Which of the following statements is/are (always) true?

(I) The system is always consistent.
(II) When the system is consistent, the general solution has exactly $n - m$ arbitrary parameters.

(A) Both (I) and (II) are true. (B) Only (I) is true.
(C) Only (II) is true. (D) Both (I) and (II) are false. (A)

Question 3 [2 marks]

Let $\mathbf{A}$ be a 3×4 non-zero matrix. Which of the following statements is/are (always) true?

(I) $\text{rank}(\mathbf{A}) \geq 1$.
(II) $\text{nullity}(\mathbf{A}) \geq 1$.
(III) $\text{nullity}(\mathbf{A}) = 0$ if and only if $\mathbf{A}$ has full rank.
(IV) $\text{nullity}(\mathbf{A}) \neq 3$.

(A) Only (I) is true. (B) Only (I) and (II) are true.
(C) Only (I) and (III) are true. (D) None of the above. (B)

Question 4 [2 marks]

Let $\mathbf{A}$ be a $m \times n$ matrix. Which of the following statements is/are (always) true?

(I) If $\text{rank}(\mathbf{A}) = n$, then $\mathbf{A}$ is invertible.
(II) If $\text{nullity}(\mathbf{A}) = 0$, then $\mathbf{A}$ is invertible.

(A) Both (I) and (II) are true. (B) Only (I) is true.
(C) Only (II) is true. (D) Both (I) and (II) are false. (D)

Question 5 [2 marks]

Let $\mathbf{A}$ and $\mathbf{B}$ be two distinct invertible matrices. Which of the following statements is/are (always) true?

(I) $\mathbf{A} - \mathbf{B}$ is invertible.
(II) $\mathbf{AB}^{-1}$ is not a diagonal matrix.

(A) Both (I) and (II) are true.
(B) Only (I) is true.
(C) Only (II) is true.
(D) Both (I) and (II) are false. (D)

Question 6 [2 marks]

Let $\mathbf{A}$ and $\mathbf{B}$ be two invertible matrices. Which of the following statements is/are (always) true?

(I) $\mathbf{A} + \mathbf{B}$ is invertible.
(II) $\mathbf{A} + \mathbf{B}$ is not invertible.

(A) Both (I) and (II) are true.
(B) Only (I) is true.
(C) Only (II) is true.
(D) Both (I) and (II) are false. (D)

Question 7 [2 marks]

Let $\mathbf{A}$ be an $m \times n$ matrix with $m \neq n$. Which of the following statements is/are true?

(I) $\mathbf{0}_n \in \text{Row}(\mathbf{A}) \cap \text{Null}(\mathbf{A})$.
(II) $\mathbf{0}_n \in \text{Col}(\mathbf{A}) \cap \text{Null}(\mathbf{A})$.
(III) If $\mathbf{x} \in \text{Row}(\mathbf{A}) \cap \text{Null}(\mathbf{A})$, then $\mathbf{x} = \mathbf{0}_n$.

(A) Only (I) is true.
(B) Only (I) and (II) are true.
(C) Only (I) and (III) are true.
(D) None of the above. (C)

Question 8 [2 marks]

Let V be a subspace of $\mathbb{R}^n$. Which of the following statements is/are (always) true?

(I) $V \cap V^{\perp} = \{\mathbf{0}\}$.
(II) $(V^{\perp})^{\perp} = V$.

(A) Both (I) and (II) are true.
(B) Only (I) is true.
(C) Only (II) is true.
(D) Both (I) and (II) are false. (A)

Question 9 [2 marks]

Which of the following matrices is/are diagonalizable?

(I) $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ (II) $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ (III) $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

(A) Only (I) is diagonalizable.
(B) Only (I) and (II) are diagonalizable.
(C) (I), (II), (III) are all diagonalizable.
(D) None of the above. (B)

Question 10 [2 marks]

Let $\mathbf{A}$ be a matrix whose characteristic polynomial is $x^2 - 3x + 2$. Which of the following statements is/are (always) true?

(I) $\mathbf{A}$ is diagonalizable.
(II) $\mathbf{A}$ is orthogonally diagonalizable.

(A) Both (I) and (II) are true.
(B) Only (I) is true.
(C) Only (II) is true.
(D) Both (I) and (II) are false. (B)

Question 11 [2 marks]

Let $\mathbf{A}$ be a matrix whose characteristic polynomial is $(x-1)^2$. Which of the following statements is/are (always) true?

(I) $\mathbf{A}$ is diagonalizable.
(II) If $\mathbf{A}$ is diagonalizable, then it is orthogonally diagonalizable.

(A) Both (I) and (II) are true.
(B) Only (I) is true.
(C) Only (II) is true.
(D) Both (I) and (II) are false. (C)

Question 12 [2 marks]

Let $\mathbf{A}$ be a square matrix of order n. Which of the following statements is/are (always) true?

(I) If $\mathbf{A}$ is diagonalizable, then it is symmetric.
(II) If $\mathbf{A}$ is orthogonally diagonalizable, then it is symmetric.
(III) If $\mathbf{A}$ is symmetric, then it is diagonalizable.
(IV) If $\mathbf{A}$ is symmetric, then it is orthogonally diagonalizable.

(A) Only (I) and (II) are true.
(B) Only (I) and (III) are true.
(C) Only (II) and (IV) are true.
(D) None of the above. (D)

Question 13 [2 marks]

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation, and S be a basis for $\mathbb{R}^n$. Which of the following statements is/are (always) true?

(I) $\ker(T) = \text{Null}([T]_S)$.
(II) $\text{R}(T) = \text{Col}([T]_S)$.

(A) Both (I) and (II) are true.
(B) Only (I) is true.
(C) Only (II) is true.
(D) Both (I) and (II) are false. (C)

Question 14 [2 marks]

Let $T, T' : \mathbb{R}^n \to \mathbb{R}^m$ be linear transformations. Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\} \subseteq \mathbb{R}^n$ such that $T(\mathbf{u}_i) = T'(\mathbf{u}_i)$ for all $i \in \{1, 2, \ldots, k\}$. Which of the following statements is/are true:

(I) If S is linearly independent, then $T(\mathbf{x}) = T'(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$.
(II) If $\text{span}(S) = \mathbb{R}^n$, then $T(\mathbf{x}) = T'(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$.

(A) Both (I) and (II) are true.
(B) Only (I) is true.
(C) Only (II) is true.
(D) Both (I) and (II) are false. (C)

Question 15 [2 marks]

Let V be a subspace of $\mathbb{R}^n$, and let $T : \mathbb{R}^n \to \mathbb{R}^n$ be the orthogonal projection onto V, i.e. for all $\mathbf{u} \in \mathbb{R}^n$, $T(\mathbf{u})$ is the orthogonal projection of $\mathbf{u}$ onto V. Which of the following statements is/are (always) true?

(I) $[T]$ is upper triangular.
(II) $[T]$ is symmetric.
(III) $[T]$ is diagonal.

(A) Only (I) is true.
(B) Only (II) is true.
(C) (I), (II), (III) are all true.
(D) None of the above. (B)

END OF PAPER

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SECTION B

INSTRUCTIONS TO CANDIDATES

1. Section B contains **SIX (6)** questions, carrying a total of **SEVENTY (70)** marks.
2. Answer **ALL** questions. Write your answers on the answer booklet provided to you.
3. The marks for each question may not be the same, and is clearly indicated at the beginning of the question.
4. This is an open-book examination, and **offline** calculators may be used. However, you should lay out systematically the various steps in the calculations.

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*Answer **all** questions.*

Question 1 [14 marks]

Let $a \neq 0$ and $b \neq 0$ be two non-zero real constants.

(a) Consider the matrix $\mathbf{A}_3 = \begin{pmatrix} a & 0 & b \\ 0 & a & 0 \\ b & 0 & a \end{pmatrix}$.

(i) For which values of a and b is $\mathbf{A}_3$ invertible? [2 marks]

(ii) When a and b satisfy the conditions in (i), compute $\mathbf{A}_3^{-1}$. [2 marks]

(b) Consider now the matrix $\mathbf{A}_4 = \begin{pmatrix} a & 0 & 0 & b \\ 0 & a & 0 & 0 \\ 0 & 0 & a & 0 \\ b & 0 & 0 & a \end{pmatrix}$.

(i) $\mathbf{A}_4$ can be transformed into an upper triangular matrix by performing a single row operation. Write down the row operation. [2 marks]

(ii) Compute $\det(\mathbf{A}_4)$. [2 marks]

(iii) For which values of a and b is $\mathbf{A}_4$ invertible? [2 marks]

(c) Consider finally the matrix $\mathbf{A}_n$ of order n generalizing $\mathbf{A}_3$ and $\mathbf{A}_4$. More precisely,

$$\mathbf{A}_n = \begin{pmatrix} a & 0 & \cdots & 0 & b \\ 0 & a & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & & \ddots & \ddots & 0 \\ b & 0 & \cdots & 0 & a \end{pmatrix}.$$

(i) Compute $\det(\mathbf{A}_n)$ for any $n \geq 2$. [2 marks]

(ii) For which values of a and b is $\mathbf{A}_n$ invertible? [2 marks]

Question 2 [14 marks]

(a) Consider the first set of vectors

$$T = \left\{ \frac{1}{5} \begin{pmatrix} 9 \\ 19 \\ 7 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ -3 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -2 \\ 7 \end{pmatrix} \right\}.$$

(i) Which pairs of vectors in T are orthogonal to each other? [2 marks]

(ii) Which vector in T has the smallest norm? [2 marks]

(b) Consider now the second set of vectors

$$S = \left\{ \begin{pmatrix} 1 \\ -1 \\ 4 \\ -2 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -2 \\ -4 \end{pmatrix} \right\}.$$

(i) Show that S is linearly independent. [2 marks]

(ii) Which vectors of T belong to span(S)? [2 marks]

(iii) For each vector of T in span(S), write down its coordinates relative to S. [2 marks]

(iv) What is the dimension of span(T)? [2 marks]

(v) What is the dimension of span$(T \cup S)$? [2 marks]

Question 3 [8 marks]

Let c and d be real constants. Find the rank and the nullity of

$$\begin{pmatrix} 2 & -4 & 4 & -2 \\ 6 & c-12 & 7 & c-6 \\ -1 & 2-d & 8 & -d+1 \end{pmatrix}$$

for all possible values of c and d.

Question 4 [8 marks]

Let $\mathbf{A} = \begin{pmatrix} 4 & 3 & -4 \\ 2 & 4 & 3 \\ 2 & 0 & 5 \\ 1 & 5 & 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 7 \\ 1 \\ -1 \\ 5 \end{pmatrix}$.

(a) Find a matrix $\mathbf{Q}$ whose columns are orthonormal and an upper triangular matrix $\mathbf{R}$ with positive diagonal entries such that $\mathbf{A} = \mathbf{QR}$. [5 marks]

(b) Find the least squares solution(s) to $\mathbf{Ax} = \mathbf{b}$. [3 marks]

Question 5 [8 marks]

Let r, s and t be real constants, and let

$$\mathbf{B} = \begin{pmatrix} r & 0 & t \\ 0 & s & 0 \\ 0 & 0 & s \end{pmatrix}.$$

Find conditions on r, s and t for $\mathbf{B}$ to be diagonalizable.

Question 6 [18 marks]

Let $\mathbf{C} = \begin{pmatrix} \sqrt{2} & -\frac{1}{\sqrt{2}} & 0 \\ -\sqrt{2} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix}$. Find a singular value decomposition for $\mathbf{C}$.

END OF PAPER